



# Synergistic representation learning via dual sample interactions for multi-view clustering

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## ARTICLE INFO

### Keywords:

Multi-view clustering  
Complementarity and consistency  
Synergistic representation learning  
Sample interaction

## ABSTRACT

In many real-world scenarios, data inherently consist of multiple views originating from diverse sources or modalities. Deep multi-view clustering has emerged as a crucial paradigm that leverages both complementarity and consistency across views to uncover latent cluster structures. However, existing methods often insufficiently exploit the complementarity between samples across different views and fail to account for feature redundancy when enforcing consistency. To address these limitations, we propose SRLS, a synergistic representation learning model driven by sample interactions for multi-view clustering. SRLS introduces a dual interaction mechanism comprising inter-sample complementarity-aware view learning and intra-sample consistency-guided feature reconstruction. Specifically, a multi-view heterogeneous graph is constructed to capture cross-view inter-sample complementarity through heterogeneous information propagation. Meanwhile, a feature reconstruction module explicitly constrains intra-sample feature interactions to reduce redundancy and highlight discriminative information. Through the synergy of these modules, the proposed end-to-end framework produces cluster-friendly representations that significantly boost clustering results. Extensive experiments on multiple real-world datasets demonstrate that SRLS outperforms state-of-the-art multi-view clustering methods, validating its effectiveness and generalizability.

## 1. Introduction

Real-world data is often inherently multi-view, comprising information from multiple sources or modalities. For example, captioned images fuse raw visual information and corresponding textual descriptions, and autonomous systems integrate data from heterogeneous sensors to develop a comprehensive understanding of complex environments. With the rapid progress in deep learning, deep multi-view clustering has emerged as a pivotal paradigm. It integrates multi-view data through deep architectures to reveal latent clustering partitions. These deep multi-view clustering approaches have been widely used in cross-modal retrieval [1], text analysis [2] and semi-supervised scenarios [3], highlighting their significant impact.

The effectiveness of deep multi-view clustering primarily stems from two inherent characteristics of multi-view data: complementarity and consistency [4]. Complementarity refers to the distinctive information uniquely captured by each view. For instance, visual modalities tend to represent fine-grained physical attributes, whereas textual data typically provides higher-level semantic context. The integration of these

heterogeneous, yet complementary cues yields a richer, more discriminative representation space. Consistency, embodies the shared latent semantics across views. Although views differ in representations, they describe the same sample and should maintain consistent structures in the underlying feature space. By integrating complementary information with consistency-driven constraints, deep multi-view clustering achieves more accurate and robust partitions in complex datasets. Despite recent advancements, existing methods still encounter two critical limitations from a clustering perspective.

First, the complementarity between samples across different views remains inadequately exploited. Existing multi-view clustering methods typically address complementarity by either fusion representations from different views [5], or constructing unified similarity graphs that aggregate diverse information from multiple views [6]. These approaches mainly focus on fusing complementary information of the same sample across views, thus enhancing the representation quality for clustering. However, from a clustering perspective, such methods often overlook the broader relationship among samples. Specifically, they pay limited attention to how the complementary information from other samples

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<https://doi.org/10.1016/j.patcog.2026.113270>

Received 24 June 2025; Received in revised form 13 January 2026; Accepted 7 February 2026

Available online 10 February 2026

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can refine the clustering of a given sample. This insufficient exploitation of inter-sample complementarity can hinder the discovery of accurate and meaningful cluster partitions.

Second, although consistency ensures cross-view alignment, it tends to overlook the presence of feature redundancy. Most current deep multi-view clustering methods rely on adversarial training or contrastive learning to enforce consistency of latent representations across multiple views. For example, adversarial approaches [7] learn view-invariant representations to facilitate effective fusion and clustering, while contrastive learning methods [8] minimize contrastive loss to enhance the consistency of representation across different views. Despite improving cross-view consistency, traditional multi-view alignment often preserves redundancy that obscures key semantic features. Therefore, a more sophisticated mechanism is needed to balance view consistency with redundancy reduction. This synergy is vital for enhancing feature discriminability and optimizing the performance of multi-view clustering.

To address the aforementioned limitations, we propose a Synergistic Representation Learning model driven by dual Sample interactions for multi-view clustering, termed SRLS. The proposed model introduces a dual interaction mechanism, which consists of an inter-sample complementarity-aware view learning and an intra-sample consistency-guided feature reconstruction. This design enables simultaneous capture of complementary information among samples across different views, while ensuring consistent feature reconstruction within each individual sample. By effectively leveraging dual sample interactions, SRLS fully exploits the rich information embedded in multi-view data, leading to enhanced representation quality and improved clustering performance. Specifically, in complementarity-aware view learning module, we construct a multi-view heterogeneous graph to capture the complementary information across different views. To better align with the clustering objective, instead of solely focusing on the cross-view information of the same sample, we further incorporate inter-sample complementary information, leveraging heterogeneous information propagation mechanisms to effectively diffuse and integrate diverse signals across neighbors and views, thereby enhancing the representation learning and facilitating the discovery of intrinsic cluster structures in the data. To further tackle the issue of feature redundancy, we introduce a consistency-guided feature reconstruction module. By explicitly modeling the intra-sample interactions among views, we enforce a constraint that encourages these interactions to align with an identity matrix, thereby effectively reducing redundancy. This reconstruction process refines the feature space to highlight both consistency and discriminative information, ultimately improving representation quality. These two modules, built upon multi-view encoding, collaboratively constitute a synergistic representation learning model that generates cluster-friendly representations in an end-to-end manner, thereby enabling effective clustering across diverse multi-view scenarios. Extensive experiments on multiple real-world datasets demonstrate that proposed model outperforms state-of-the-art multi-view clustering methods.

## 2. Related work

Multi-view clustering [9] aims to achieve enhanced clustering performance by integrating information from different views. With the growing diversity of data forms, deep multi-view clustering has emerged as one of the most popular tasks in data mining with its capability in representation learning [10]. This paper divides existing approaches into two categories based on how multi-view data characteristics are exploited: one based on consistency, focusing on ensuring alignment among multiple views; the other based on complementarity, aiming to integrate diverse information from each view.

With the rapid development of deep self-supervised methods such as contrastive learning, the majority of multi-view clustering approaches are gradually based on consistency. These methods focus on maximizing the consensus among different views to ensure alignment across views. The work in [11] used an attention mechanism to achieve consistent rep-

resentations. MFLVC [8] is designed through multi-level feature learning for contrastive multi-view clustering to improve clustering performance. CVCL [12] learns invariant representations by contrasting cluster assignments across views. PGCL-DCL [13] combines pseudo-label guided contrastive learning and dual correlation learning to enhance discriminative multiview clustering. PDMC-RCL [14] employs dual-level reliable contrastive learning to adaptively weight heterogeneous modality pairs for robust multi-modal clustering. Despite these advancements, challenges such as incomplete multi-view data [15] persist. IMVC-TGR [16] leverages graph refinement and tensor constraints to eliminate noise and exploit high-order inter-view consistency for effective incomplete clustering. BWIC-TIMC [17] integrates a dual tensor constraint with adaptive graph fusion to bridge missing views by balancing between-view and within-view information.

As for complementarity, following the previous work [5], the use of either within-view neighbor information or cross-view diversity information for multi-view clustering is referred to as complementarity-based methods. We firstly discuss multi-view clustering methods based on neighborhood or structure complementarity. DIVIDE [18] is a robust contrastive multi-view clustering method that uses high-order random walks to consider the sample neighborhood. AdaM [19] leverages structure-enhanced contrastive learning and adaptive consistency alignment to learn robust consensus representations for multi-view clustering. CBF-CGF [20] integrates graph filters with bidirectional fusion to exploit the complementarity of high-order neighborhood information for robust multi-view consistency embedding. TGNN [21] disentangles determinative factors from incidental noise to aggregate task-oriented semantics, leveraging cross-view complementarity for discriminative multi-view representation learning. CCARL [22] integrates complementary information across views via an attention mechanism to learn global context-aware representations. Additionally, some approaches also consider the complementarity across different views. SURER [23] refines view-specific attribute graphs and integrates them into a unified heterogeneous graph, which is used by a graph neural network to learn a representation for clustering. CTCC [24] is a structural multi-view clustering framework, that integrates consistent and complementary information from multiple views by maximizing mutual information and utilizing cross-view topology based on optimal transport distance. VMC-CD [25] effectively balances consistency and diversity in multi-view clustering through dual contrastive learning.

## 3. Proposed method

The multi-view clustering task is to partition a collection of  $N$  samples, represents as  $\{\mathbf{X}^v\}_{v=1}^V = \{(\mathbf{x}_1^v, \dots, \mathbf{x}_i^v, \dots, \mathbf{x}_I^v)\}_{i=1}^N$ , into  $C$  distinct clusters. Here,  $v$  represents the  $v$ th view and  $V$  denotes the number of views. We aim to construct a synergistic learning paradigm that integrates multi-view consistency and complementarity. This design ensures the extraction of cluster-prescriptive representations, thereby enhancing the overall precision of the clustering process.

The overall architecture of proposed SRLS is illustrated in Fig. 1. It consists of two main phases: (1) Multi-view encoding, which extracts informative representations from each view; (2) Synergistic representation learning via dual sample interactions through two modules: the complementarity-aware view learning module for leveraging inter-sample neighbor relationships, and the consistency-guided feature reconstruction module for encouraging intra-sample feature invariance and reducing redundancy. The outputs of these modules are jointly optimized using a unified objective function tailored for efficient multi-view clustering.

### 3.1. Multi-view encoding

The multi-view encoding serves as the backbone of the proposed SRLS, enabling the extraction of representations to facilitate subsequent modules. It consists of a series of Siamese within-view encoders and

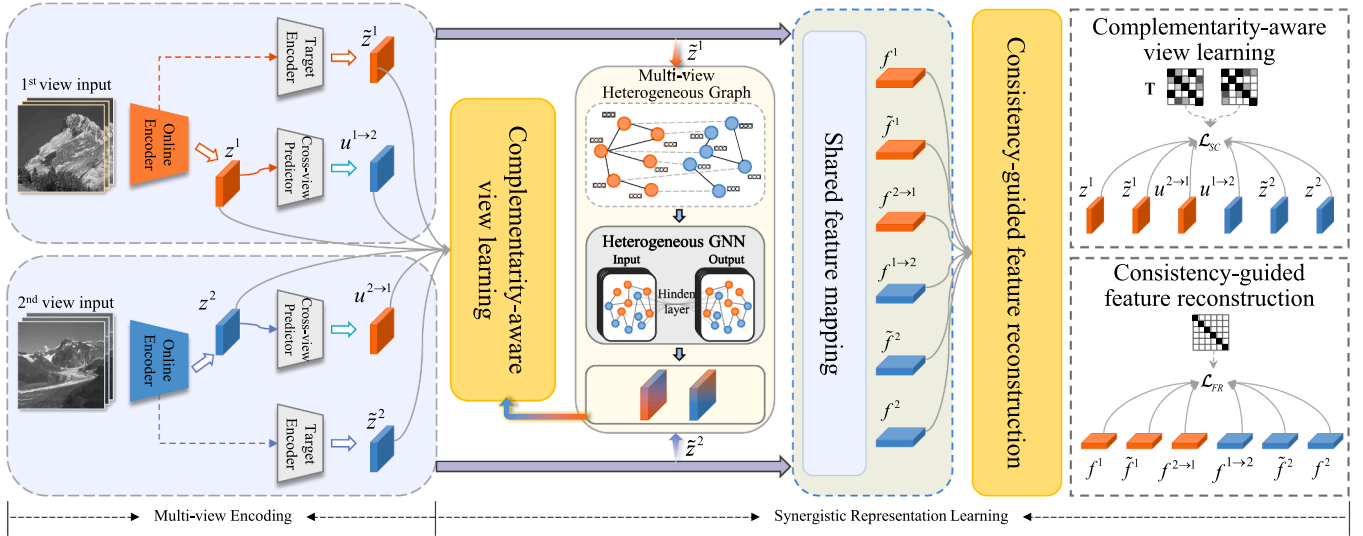


Fig. 1. The architecture of the proposed SRLS.

cross-view predictors. Each Siamese within-view encoder is employed to learn representations from individual view. And the cross-view predictors project representations from one view to other views by applying two-layer nonlinear fully connected networks.

Let  $(\psi_e^v(\cdot); \psi_e^v(\cdot))$  be the parameterized Siamese within-view encoders for the  $v$ th view. We can get the encoding representation  $\mathbf{Z}^v$  and  $\tilde{\mathbf{Z}}^v$  as follows:

$$\mathbf{Z}^v = \psi_e^v(\mathbf{X}^v); \quad \tilde{\mathbf{Z}}^v = \psi_e^v(\mathbf{X}^v). \quad (1)$$

Specifically,  $\psi_e^v$  is formulated as a slowly progressing version of  $\psi_e^v$  through a momentum update:  $\psi_e^v \leftarrow m\psi_e^v + (1-m)\psi_e^v$ . By decoupling the parameter updates of the two branches, this strategy mitigates the risk of gradient collapse and enhances training robustness.

Let  $\psi_p^{v \rightarrow \bar{v}}(\cdot)$  be the parameterized cross-view predictor for the view  $v$  to view  $\bar{v}$ , then the encoding representation can be obtained by:

$$\mathbf{U}^{v \rightarrow \bar{v}} = \psi_p^{v \rightarrow \bar{v}}(\mathbf{Z}^v) \quad (2)$$

### 3.2. Synergistic representation learning

**The complementarity-aware view learning.** To better leverage sample complementarity, we propose a complementarity-aware view learning approach. This method constructs a multi-view heterogeneous graph that simultaneously captures both within-view and cross-view complementary relationships.

Formally, for within-view connectivity, an edge is instantiated between  $\tilde{\mathbf{z}}_i^v$  and  $\tilde{\mathbf{z}}_j^v$  if  $\mathbf{x}_j^v$  resides in the neighborhood of  $\mathbf{x}_i^v$ , capturing view-specific manifold information. For cross-view interaction, we link the multi-view representations of the same sample, denoted as  $\tilde{\mathbf{z}}_i^v$  and  $\tilde{\mathbf{z}}_i^{\bar{v}}$  ( $\bar{v} \neq v$ ), to enforce cross-perspective consistency. Furthermore, we generalize these links to cross-view neighbors, enabling the model to capture complex, non-linear complementary relationships beyond simple instance-wise alignment. Specifically, for a sample  $i$  in view  $v$ , the cross-view neighbor edges connect to the  $k$ -nearest neighbors of sample  $i$  identified in other views  $\tau \neq v$ . This mechanism enables the model to leverage topological structures specific to other views, effectively complementing the information of the current view. The heterogeneous graph neural network iteratively updates the view representation  $\mathbf{h}_i^v \in \mathbf{H}^v$  with the message propagation as follows:

$$\mathbf{h}_i^v = \mathbf{W}_1 \tilde{\mathbf{z}}_i^v + \mathbf{W}_2 \frac{1}{k} \sum_{j=1}^k \tilde{\mathbf{z}}_j^v + \mathbf{W}_3 \frac{1}{V-1} \sum_{\tau \neq v} \frac{1}{k} \sum_{j=1}^k \tilde{\mathbf{z}}_j^\tau. \quad (3)$$

Mathematically, the update rule in Eq. (3) decomposes the heterogeneous information propagation into three distinct components, gov-

erned by learnable weight matrices  $\mathbf{W}_1$ ,  $\mathbf{W}_2$ , and  $\mathbf{W}_3$ . The first term serves as a self-preservation mechanism that transforms the node's own representation to retain unique view-specific features. The second term captures the local geometric structure by aggregating information from neighbors within the same view, ensuring smoothness consistent with the within-view topology. Crucially, distinguishing our approach from conventional methods that rely solely on same-sample alignment, the third term introduces an explicit cross-view inter-sample interaction. By aggregating neighbor information from all other views, this mechanism allows the structural complementarity to directly refine the target view's representation, thereby realizing a comprehensive heterogeneous information propagation.

Furthermore, in order to incorporate neighboring complementarity in own view and other views into view representation, we also utilize the output of Eq. (3) as inputs and iteratively repeat this propagation process in our experiments. Formally, we denote the rule in Eq. (3) as a mapping  $\mathcal{G}(\cdot)$  and define the iterative update as  $\mathbf{H}^{(l+1)} = \mathcal{G}(\mathbf{H}^{(l)})$ , where  $\mathbf{H}^{(0)} = \{\tilde{\mathbf{z}}_i^v\}$ . Stacking  $L$  such layers allows the heterogeneous graph neural network to aggregate higher-order neighborhood information, thereby enhancing the discriminative power of the learned representations.

Then, we construct a fully-connected affinity graph for the in-batch samples by regarding the representations as nodes and defining the edge weights with the heat kernel similarity  $a_{ij}^v \in \mathbf{A}^v$ , and  $a_{ij}^v = \exp\left(-\frac{\|\mathbf{h}_i^v - \mathbf{h}_j^v\|^2}{\gamma}\right)$ . Here  $\gamma$  is the scale parameter that controls the smoothness of the similarity, which is set to 0.1 in this paper.

To integrate the complementary insights embedded in different views for enhanced clustering performance, the  $\mathcal{L}_{VL}$  is constructed as:

$$\mathcal{L}_{VL} = \sum_v J_s(\mathbf{S}^v, \sigma(\mathbf{Z}^v, \tilde{\mathbf{Z}}^v)) + \sum_{v, \bar{v}} J_s(\mathbf{S}^v, \sigma(\mathbf{U}^{v \rightarrow \bar{v}}, \tilde{\mathbf{Z}}^v)) \quad (4)$$

$$\mathbf{S}^v = \alpha \mathbf{I}_M + (1 - \alpha) \mathbf{M}^{v(l)} \quad (5)$$

where  $J_s(\cdot, \cdot)$  indicates the cross entropy,  $\mathbf{S}^v \in \mathbb{R}^{n \times n}$  is the pseudo target that contains the neighborhood complementarity of samples across multiple views, and  $\sigma(\cdot, \cdot)$  is the pairwise similarity.  $\mathbf{I}_M$  denotes the identity matrix of the same order as matrix  $\mathbf{M}^v$ . And  $\mathbf{M}^v$  is obtained by normalizing  $\mathbf{A}^v$ ;  $(\cdot)^{(l)}$  denotes the  $l$ th order. In our experiment,  $\alpha$  is set to 0.5.

**The consistency-guided feature reconstruction.** To tackle the issue of cross-view feature redundancy, we propose a consistency-guided feature reconstruction approach. We employ a shared feature mapper designed to project diverse view representations into a common feature space,

ensuring both effective embedding transformation and cross-view alignment.

The different types of view representation are fed into the shared feature mapper  $\psi_\zeta(\cdot)$  with trainable parameters  $\zeta$ . Taking  $\tilde{\mathbf{z}}_i^v \in \tilde{\mathbf{Z}}^v$  as an illustrative example, its corresponding feature embedding is expressed as:

$$\tilde{\mathbf{f}}_i^v = \psi_\zeta(\tilde{\mathbf{z}}_i^v) \quad (6)$$

where  $\tilde{\mathbf{f}}_i^v \in \mathbb{R}^{d_F}$  and  $d_F$  is the output dimension of  $\psi_\zeta(\cdot)$ .

To ensure view-invariant representations and reduce redundancy via feature decorrelation, we adopt a loss function defined as follows:

$$l(\psi_\zeta(\mathbf{Z}^v), \psi_\zeta(\tilde{\mathbf{Z}}^v); r) = \mathbf{I}(\mathbf{F}^v, \tilde{\mathbf{F}}^v; r) \quad (7)$$

$$= \sum_{m,n}^{d_F} [(1 - c_{mn}^{(v,v)}(r))^2 \cdot \delta_{mn} + \lambda (c_{mn}^{(v,v)}(r))^2 \cdot (1 - \delta_{mn})] \quad (8)$$

where  $\delta_{mn}$  denotes the Kronecker delta function. The hyperparameter  $r \in (0, 1]$  represents the sampling ratio, governing the proportion of samples within a batch that undergo feature reconstruction. Furthermore,  $\lambda > 0$  serves as a penalty coefficient to balance the contribution of the invariance term against the redundancy reduction term. While  $r$  governs the computational scope within each batch,  $\lambda$  and  $\delta_{mn}$  synergistically shape the geometric structure of the feature space by promoting cross-view consistency and mitigating information redundancy. Following [26], we set  $\lambda = 0.0051$  without any further discussion here. For further details, please refer to the original manuscript.  $c_{mn}^{(v,v)}(r)$  refers to the cross-correlation indicator computed between the outputs of feature mapper along the batch dimensions as follows:

$$c_{mn}^{(v,v)}(r) \triangleq \frac{\sum_{b=1}^{\lceil br \rceil} \mathbf{f}_{b,m}^v \tilde{\mathbf{f}}_{b,n}^v}{\sqrt{\sum_{b=1}^{\lceil br \rceil} (\mathbf{f}_{b,m}^v)^2} \sqrt{\sum_{b=1}^{\lceil br \rceil} (\tilde{\mathbf{f}}_{b,n}^v)^2}} \quad (9)$$

where  $b$  indexes the batch samples, and  $m$  and  $n$  index the dimensions of the feature embeddings.  $\lceil \cdot \rceil$  denotes the ceiling operation. In addition,  $\mathbf{f}_{b,m}^v$  and  $\tilde{\mathbf{f}}_{b,n}^v$  are assumed to be mean-centered along the batch dimension, such that each unit has a mean output 0 over the batch. In our objective function (Eq. (8)), the second term  $(c_{mn}^{(v,v)}(r))^2 \cdot (1 - \delta_{mn})$  explicitly targets the off-diagonal elements of the cross-correlation matrix. Mathematically, minimizing these off-diagonal terms toward zero forces the feature dimensions to be statistically uncorrelated. According to the redundancy reduction principle established in [26], decorrelating the output variables minimizes the mutual information between feature channels, thereby ensuring that the representation contains non-redundant information. Consequently, forcing the cross-correlation matrix toward the identity matrix  $\mathbf{I}$  simultaneously achieves two goals: the diagonal terms guarantee view-consistency, while the off-diagonal terms minimize redundancy.

Therefore,  $\mathcal{L}_{FR}$  is formed as follows:

$$\mathcal{L}_{FR} = \sum_v l(\psi_\zeta(\mathbf{Z}^v), \psi_\zeta(\tilde{\mathbf{Z}}^v); r) + \sum_{v,\tilde{v}} l(\psi_\zeta(\mathbf{U}^{v \rightarrow \tilde{v}}), \psi_\zeta(\tilde{\mathbf{Z}}^{\tilde{v}}); r) \quad (10)$$

**The synergistic representation learning loss.** Combining the above two loss terms, the overall objective function of proposed model is:

$$\mathcal{L} = \mathcal{L}_{VL} + \beta \mathcal{L}_{FR} \quad (11)$$

where  $\mathcal{L}_{VL}$  is a complementarity-aware view learning loss designed to capture the inter-sample complementarity across views, and  $\mathcal{L}_{FR}$  is a consistency-guided feature reconstruction loss, aimed at mitigating feature redundancy.  $\beta$  is a controlling parameter which balances the contribution of  $\mathcal{L}_{VL}$  and  $\mathcal{L}_{FR}$ .

This joint optimization enables us to better exploit the inter-sample complementary information and intra-sample consistency of features inherent in multi-view data, thereby providing enhanced support for downstream clustering tasks. In our experiments, k-means [27] is employed on the concatenated view representations for clustering to illustrate the effectiveness of the proposed synergistic representation learning model. Its details are shown in Algorithm 1.

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#### Algorithm 1 The proposed SRLS model.

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**Require:** Multi-view dataset  $\{\mathbf{X}^v\}_{v=1}^V$  with  $N$  samples, number of clusters  $C$ , maximum iterations  $T$

**Ensure:** Cluster assignments  $\{y_i\}_{i=1}^N$

- 1: **while**  $t < T$  **do**
- 2: Initialize the Siamese view representations via Eq. (1) and cross-view predictions via Eq. (2)
- 3: //complementarity-aware view learning
- 4: Learn the inter-sample complementarity through multi-view heterogeneous GCN, and construct a pseudo target  $\mathbf{S}^v$  for each view
- 5: Calculate the complementarity-aware view learning loss via Eq. (4)
- 6: //consistency-guided feature reconstruction
- 7: Get the feature embedding  $\tilde{\mathbf{F}}^v$  by Eq. (6) and learn the intra-sample consistency through cross-correlation
- 8: Calculate the consistency-guided feature reconstruction loss via Eq. (10)
- 9: //model optimization
- 10: Optimize the entire model via synergistic representation learning loss (Eq. (11))
- 11:  $t \leftarrow t + 1$
- 12: **end while**
- 13: Apply k-means to the concatenated view representations
- 14: **return**  $\{y_i\}_{i=1}^N$

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### 3.3. Computational complexity analysis

The computational complexity of SRLS per iteration comprises three parts: (i) multi-view encoding:  $O(V \cdot B \cdot d^2)$ , (ii) complementarity-aware view learning (propagation and affinity matrix):  $O(V \cdot B \cdot k \cdot d + V \cdot B^2 \cdot d)$ , and (iii) consistency-guided feature reconstruction:  $O(V^2 \cdot B \cdot d_F + V^2 \cdot d_F^2)$ . The total complexity is summarized as  $O(V \cdot B \cdot d^2 + V \cdot B^2 + V^2(B \cdot d_F + d_F^2))$ , where  $B$ ,  $V$ ,  $d$ , and  $d_F$  denote the batch size, views, and dimensions, respectively. As  $B \ll N$ , the overall training complexity scales linearly with the sample size  $N$ , ensuring high efficiency for large-scale datasets.

## 4. Experiment

In this section, we implement experiments to verify the effectiveness of the proposed SRLS through answering the following questions: (Q1) Does SRLS outperform state-of-the-art multi-view clustering methods? (Q2) Does each component of SRLS contribute to the overall performance? (Q3) How do the hyper-parameters impact the clustering performance of SRLS? (Q4) What is the clustering structure revealed by SRLS? (Q5) Could SRLS handle multi-view clustering with missing views? (Q6) Does SRLS maintain competitive computational efficiency?

### 4.1. Experimental setup

To evaluate the effectiveness of the proposed SRLS, we carried out extensive experiments on five benchmark datasets: Scene15 [15], LandUse21 [28], CUB [29], and Yale [30]. Additionally, we include Reuters, a large-scale benchmark dataset, listed in [10]. Specifically, we adopt the 18,785-sample version as specified in [15]. A succinct overview is presented in Table 1.

The experiments are conducted utilizing the following hardware setup: Intel Xeon Platinum 8370C CPU and NVIDIA RTX A6000 GPU. In addition, the PyTorch platform is used for all experiments. In the case of our SRLS, a batch size of 1024 is used across all datasets. We used a fully connected network (FCN) of 4 layers with ReLU activation to construct the Siamese encoder within view, using an update momentum of 0.99, and the cross-view predictor is an FCN of 2 layers. The



**Table 1**  
Overview of benchmark datasets.

| Dataset   | Samples | Clusters | Dimensions |
|-----------|---------|----------|------------|
| Scene15   | 4485    | 15       | 20/59      |
| LandUse21 | 2100    | 21       | 59/40      |
| CUB       | 600     | 10       | 1024/300   |
| Yale      | 165     | 15       | 4096/3304  |
| Reuters   | 18,758  | 6        | 10/10      |

weight matrices  $\mathbf{W}_1$ ,  $\mathbf{W}_2$ , and  $\mathbf{W}_3$  in Eq. (3) (implemented as linear layers) are initialized using the Kaiming Uniform Initialization. For the optimization, we employed the Adam optimizer with default momentum parameters (0.9, 0.999). To ensure optimal performance, the learning rate was dataset-specific:  $5 \times 10^{-4}$  for Scene15, Yale, and Reuters, while for LandUse21 and CUB, it is set to  $5 \times 10^{-3}$  and  $1 \times 10^{-3}$ , respectively. To warm up, we set the target  $\mathbf{M}^v$  in Eq. (5) as an identity matrix  $\mathbf{I}$  for the first 100 epochs and then adopt the  $\mathbf{M}^v$  in the remaining epochs. In the experiments, we conducted 5 runs and reported the average of the experimental results in the main comparisons.

For a comprehensive analysis, we used three widely used clustering metrics to evaluate the performance, including clustering Accuracy (ACC), Normalized Mutual Information (NMI) and Adjusted Rand Index (ARI). A higher value of these metrics indicates a better clustering performance.

#### 4.2. Comparisons with state of the arts (Q1)

The proposed SRLS is benchmarked against existing deep multi-view clustering methods. Specifically, these compared clustering algorithms can be broadly categorized into two groups: redconsistency-based multi-view clustering methods (Completer, ProImp, CVCL, MFLVC, LDMRL-v, LDMRL-e and 3MC) and complementarity-based multi-view clustering methods (SDSNE, SURER, ICMVC, MAGA, GCFAGG and DIVIDE). It is worth mentioning that for all comparison methods, we used the parameters recommended in the original papers and reproduced the experiments under the same experimental conditions. The results of the comparative experiments are all shown in Table 2, and we have the following findings:

(1) The experimental results demonstrate that the proposed SRLS consistently outperforms existing methods in terms of overall clustering performance. This superior performance can be attributed to the effective integration of both complementarity-aware view learning and consistency-guided feature reconstruction across multiple views. The performance of existing multi-view clustering methods varies across datasets with different types. For example, DIVIDE, which utilizes inter-sample complementarity, performs similarly to the proposed SRLS on the LandUse21 dataset, whereas ProImp, which emphasizes intra-sample view consistency, outperforms on the Reuters dataset. Although proposed model utilizes only the straightforward k-means algorithm, it achieves the best or second-best clustering performance. This result substantiates its effectiveness in capturing clustering-friendly multi-view representations and producing more discriminative cluster partitions.

(2) Unlike most existing consistency-based multi-view clustering methods, the proposed model discards the alignment of view representations for the same sample. Instead, it achieves intra-sample feature alignment through feature reconstruction, thereby enabling semantic matching across views. This not only reduces feature redundancy but also enhances discriminability.

(3) Compared with existing complementarity-based deep multi-view clustering methods, such as SURER, which focuses on multi-view complementarity, the proposed model not only leverages conventional cross-view complementary information, but also further explores sample relationships in light of the clustering objective. By integrating inter-sample neighborhood information across multiple views, the resulting view representations exhibit superior cluster representativeness.

(4) To further evaluate the impact of key training settings in SRLS, we introduce two variants: SRLS without the Warmup training (SRLS w/o Wp) and SRLS without the Momentum-updated target Encoder (SRLS w/o ME). Results in Table 2 show that the SRLS model outperforms all variants. Specifically, SRLS w/o ME suffers the largest performance drop (e.g., on CUB and Reuters), proving that the momentum target encoder is critical for training stability. SRLS w/o Wp also shows lower accuracy on datasets like Scene15, indicating that the Warmup strategy effectively guides the model to a better optimization starting point. Overall, the results confirm that both components are essential for achieving superior performance.

#### 4.3. Ablation studies and method analysis (Q2)

*Effectiveness of each loss term.* To investigate the contribution of the two terms in the proposed synergistic representation learning loss in Eq. (11), we conduct ablation experiments on  $\mathcal{L}_{VL}$  and  $\mathcal{L}_{FR}$  across all datasets, as presented in Table 3. It is observed that using  $\mathcal{L}_{VL}$  alone provides a modest improvement over the baseline but is not sufficient to fully exploit the multi-view information, as reflected by relatively lower scores in all three metrics.

The best clustering performance is consistently achieved when both  $\mathcal{L}_{VL}$  and  $\mathcal{L}_{FR}$  are jointly applied. This demonstrates that the two loss terms are complementary:  $\mathcal{L}_{VL}$  enhances complementarity-aware view learning by focusing on inter-sample interactions, while  $\mathcal{L}_{FR}$  facilitates feature reconstruction to preserve intra-sample view consistency. Their combination significantly improves ACC, NMI, and ARI on all datasets. For instance, on the CUB dataset, the full model achieves 83.10% ACC, 76.56% NMI, and 68.48% ARI, outperforming other variants by a noticeable margin. Similar trends are observed on Scene15, LandUse21, Yale, and Reuters, validating the effectiveness of the proposed dual sample interaction representation learning.

*Effectiveness of the complementary information across views.* To thoroughly investigate the contribution of each component in complementarity-aware view learning, we conduct ablation experiments under four variants: (1) using only  $\mathbf{W}_1$ , which encodes the individual view representation  $\tilde{\mathbf{z}}_i^v$ ; (2) using  $\mathbf{W}_1$  and  $\mathbf{W}_2$ , in which  $\mathbf{W}_2$  incorporates the within view complementarity by averaging over the  $k$  nearest neighbors in the same view; (3) using  $\mathbf{W}_1$  and  $\mathbf{W}_3$ , in which  $\mathbf{W}_3$  introduces cross-view information by aggregating the neighbors from other views; and (4) combining all three components  $\mathbf{W}_1$ ,  $\mathbf{W}_2$ , and  $\mathbf{W}_3$  to fully exploit both within-view and cross-view neighborhood complementarity. These settings allow us to evaluate the individual and joint effectiveness of different cues in enhancing the learned representation  $\mathbf{h}_i^v$ .

Table 4 examines the effect of incorporating complementary information through the weights in Eq. (3) on the performance of the clustering of five datasets. When only  $\mathbf{W}_1$  is used, all metrics (ACC, NMI, ARI) reach their lowest values, reflecting the limitations of the data itself. Introducing a second weight ( $\mathbf{W}_1$  &  $\mathbf{W}_2$  or  $\mathbf{W}_1$  &  $\mathbf{W}_3$ ) consistently improves performance on most datasets, demonstrating that complementary fusion captures richer cluster structure. Finally, combining three weights ( $\mathbf{W}_1$  &  $\mathbf{W}_2$  &  $\mathbf{W}_3$ ) yields the highest—or near highest—performances on every benchmark, confirming that each additional weight contributes complementary information and produces the most discriminative view representations.

#### 4.4. Influence of hyper-parameters (Q3)

A series of experiments were conducted on diverse datasets to systematically examine the effect of key hyper-parameters.

*Influence of  $\beta$ .* Investigate the impact of the trade-off parameter  $\beta$  in Eq. (11), which balances the complementarity-aware view learning loss  $\mathcal{L}_{VL}$  and the consistency-guided feature reconstruction loss  $\mathcal{L}_{FR}$ , we

**Table 2**

Clustering performance across five multi-view benchmark datasets. The most exceptional results are marked in **bold**, and the second-best values are underlined. The “-” means we cannot get clustering results on the dataset because the large-scale dataset occupies many computation resources.

| Methods       | Scene15       |               |               | LandUse21     |               |               | CUB           |               |               | Yale          |               |               | Reuters       |               |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|               | ACC           | NMI           | ARI           | ACC           | NMI           | ARI           | ACC           | NMI           | ARI           | ACC           | NMI           | ARI           | ACC           | NMI           | ARI           |
| Completer[31] | 0.3970        | 0.4225        | 0.2348        | 0.2569        | 0.3255        | 0.1386        | 0.5850        | 0.7160        | 0.5389        | 0.5727        | 0.6148        | 0.2942        | 0.3995        | 0.2223        | 0.0732        |
| ProImp[32]    | 0.4499        | 0.4477        | 0.2738        | 0.2414        | 0.2843        | 0.1121        | 0.8244        | <u>0.7653</u> | <u>0.6749</u> | 0.6000        | 0.6527        | 0.4280        | <u>0.5415</u> | <u>0.3815</u> | <u>0.3183</u> |
| CVCL[12]      | 0.4216        | 0.4338        | 0.2523        | 0.2295        | 0.2465        | 0.0963        | 0.8019        | 0.7180        | 0.6291        | 0.4357        | 0.5597        | 0.2938        | 0.4061        | 0.1592        | 0.1312        |
| MFLVC[8]      | 0.4018        | 0.4238        | 0.2378        | 0.2119        | 0.2658        | 0.1034        | 0.7450        | 0.7513        | 0.6198        | 0.3394        | 0.4057        | 0.1222        | 0.5190        | 0.3458        | 0.2856        |
| LDMRL-v[33]   | 0.3824        | 0.4165        | 0.2180        | 0.1957        | 0.2156        | 0.0728        | 0.6167        | 0.6062        | 0.4485        | 0.4848        | 0.5131        | 0.2432        | 0.4587        | 0.3162        | 0.2437        |
| LDMRL-e[33]   | 0.3291        | 0.3792        | 0.1891        | 0.1776        | 0.2221        | 0.0513        | 0.5033        | 0.5490        | 0.3623        | 0.5152        | 0.5522        | 0.2993        | 0.4651        | 0.2832        | 0.2341        |
| 3MC[34]       | 0.3525        | 0.3930        | 0.2002        | 0.2224        | 0.258         | 0.0955        | 0.7700        | 0.7367        | 0.6355        | 0.4545        | 0.5175        | 0.2598        | 0.4851        | 0.2679        | 0.2329        |
| SDSNE[35]     | 0.4252        | 0.4391        | 0.2358        | 0.2510        | 0.3303        | 0.0859        | 0.7467        | 0.7627        | 0.6249        | 0.5939        | 0.6404        | 0.4222        | -             | -             | -             |
| SURER[23]     | 0.4060        | 0.3868        | 0.2013        | 0.2624        | 0.3036        | 0.1129        | <u>0.8300</u> | 0.7647        | 0.6670        | <u>0.6182</u> | <u>0.6592</u> | <u>0.4409</u> | 0.5592        | 0.3700        | 0.3009        |
| ICMVC[36]     | 0.4276        | 0.4193        | 0.2487        | 0.2744        | 0.3227        | 0.1377        | 0.7633        | 0.7357        | 0.6339        | 0.4545        | 0.5848        | 0.3574        | 0.4907        | 0.3398        | 0.2615        |
| MAGA[37]      | 0.3561        | 0.3973        | 0.2076        | 0.2214        | 0.2418        | 0.0855        | 0.6700        | 0.7087        | 0.5643        | 0.5141        | 0.3906        | 0.3039        | 0.4497        | 0.3045        | 0.2276        |
| GCFAGG[6]     | 0.3554        | 0.3813        | 0.2155        | 0.2748        | 0.3093        | 0.1374        | 0.7967        | 0.7493        | 0.6444        | 0.3879        | 0.4894        | 0.2049        | 0.4883        | 0.3769        | 0.2290        |
| DIVIDE[18]    | <u>0.4877</u> | <u>0.4861</u> | <u>0.3133</u> | <u>0.3175</u> | <b>0.3998</b> | <u>0.1778</u> | 0.7537        | 0.7489        | 0.6483        | 0.6145        | 0.6398        | 0.4167        | 0.4760        | 0.3224        | 0.2176        |
| SRLS          | <b>0.5068</b> | <b>0.5100</b> | <b>0.3376</b> | <b>0.3287</b> | <u>0.3942</u> | <b>0.1818</b> | <b>0.8310</b> | <b>0.7656</b> | <b>0.6848</b> | <b>0.6558</b> | <b>0.6669</b> | <b>0.4532</b> | 0.5855        | <b>0.3831</b> | <b>0.3253</b> |
| SRLS w/o Wp   | 0.5005        | 0.5100        | 0.3341        | 0.3090        | 0.3875        | 0.1720        | 0.8207        | 0.7599        | 0.6755        | 0.6558        | 0.6669        | 0.4532        | 0.5855        | 0.3831        | 0.3253        |
| SRLS w/o ME   | 0.4577        | 0.4754        | 0.2827        | 0.2981        | 0.3523        | 0.1576        | 0.7120        | 0.6790        | 0.5464        | 0.5976        | 0.6337        | 0.3938        | 0.4929        | 0.3363        | 0.2495        |

**Table 3**

Ablation studies on two losses.

| $\ast \mathcal{L}_{VL}$ | $\mathcal{L}_{FR}$ | Scene15       |               |               | LandUse21     |               |               | CUB           |               |               | Yale          |               |               | Reuters       |               |               |
|-------------------------|--------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|                         |                    | ACC           | NMI           | ARI           | ACC           | NMI           | ARI           | ACC           | NMI           | ARI           | ACC           | NMI           | ARI           | ACC           | NMI           | ARI           |
| ✓                       |                    | 0.4716        | 0.4769        | 0.3008        | 0.3188        | <b>0.3949</b> | 0.1760        | 0.7623        | <b>0.7737</b> | 0.6534        | 0.6436        | 0.6423        | 0.4262        | 0.4133        | 0.2203        | 0.1354        |
|                         | ✓                  | 0.4943        | 0.5054        | 0.3273        | 0.3100        | 0.3751        | 0.1612        | 0.8130        | 0.7512        | 0.6644        | 0.6315        | 0.6384        | 0.4076        | 0.5051        | 0.3486        | 0.2325        |
| ✓                       | ✓                  | <b>0.5068</b> | <b>0.5100</b> | <b>0.3376</b> | <b>0.3287</b> | 0.3942        | <b>0.1818</b> | <b>0.8310</b> | 0.7656        | <b>0.6848</b> | <b>0.6558</b> | <b>0.6669</b> | <b>0.4532</b> | <b>0.5855</b> | <b>0.3831</b> | <b>0.3253</b> |

**Table 4**

Influence of the complementary information across views.

| Method                        | Scene15 |        |        | LandUse21 |        |        | CUB    |        |        | Yale   |        |        | Reuters |        |        |
|-------------------------------|---------|--------|--------|-----------|--------|--------|--------|--------|--------|--------|--------|--------|---------|--------|--------|
|                               | ACC     | NMI    | ARI    | ACC       | NMI    | ARI    | ACC    | NMI    | ARI    | ACC    | NMI    | ARI    | ACC     | NMI    | ARI    |
| $W_1$                         | 0.5010  | 0.5074 | 0.3335 | 0.3262    | 0.3927 | 0.1783 | 0.8193 | 0.7540 | 0.6693 | 0.6521 | 0.6636 | 0.4471 | 0.5992  | 0.3931 | 0.3354 |
| $W_1 \& W_2$                  | 0.5085  | 0.5118 | 0.3410 | 0.3249    | 0.3861 | 0.1720 | 0.8087 | 0.7614 | 0.6686 | 0.6509 | 0.6606 | 0.4424 | 0.5945  | 0.3942 | 0.3399 |
| $W_1 \& W_3$                  | 0.4881  | 0.5071 | 0.3309 | 0.3186    | 0.3879 | 0.1733 | 0.8090 | 0.7505 | 0.6575 | 0.6558 | 0.6669 | 0.4532 | 0.5887  | 0.3735 | 0.3145 |
| $W_1 \& W_2 \& W_3$ (Eq. (3)) | 0.5068  | 0.5100 | 0.3376 | 0.3287    | 0.3942 | 0.1818 | 0.8310 | 0.7656 | 0.6848 | 0.6558 | 0.6669 | 0.4532 | 0.5855  | 0.3831 | 0.3253 |

conduct a sensitivity analysis by varying  $\beta$  from 0.001 to 10. Fig. 2 reports the average clustering performance between different values  $\beta$  in the Scene15 and CUB datasets, with error bars denoting standard deviations.

Our sensitivity analysis across different datasets (Fig. 2) confirms this trade-off. On the Scene15 dataset, performance peaks within a narrow range [0.001, 0.01] and declines as  $\beta$  increases, with ARI exhibiting the most significant sensitivity. This suggests that for datasets with high intra-cluster variability and visual diversity,  $\mathcal{L}_{VL}$  plays a dominant role, and excessive emphasis on consistency can be counterproductive. In contrast, the performance on the CUB dataset remains comparatively stable across a broader range of  $\beta$ , indicating higher robustness in scenarios with more consistent semantics. Based on these observations, we identified  $\beta = 0.001$  as the optimal trade-off point, providing a necessary regularization effect to ensure feature purity without overshadowing the rich complementarity.

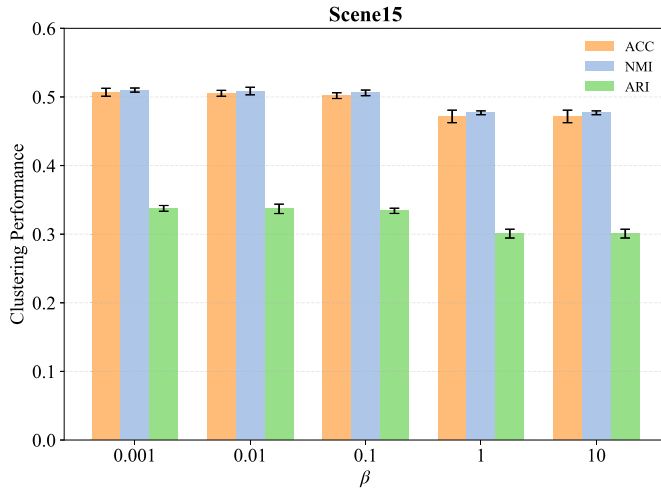
**Influence of  $k$ .** In the complementarity-aware view learning module, the parameter  $k$  determines the number of nearest neighbors for each sample, thus influencing local information. To evaluate its effect, we varied  $k$  from 5 to 25 in increments of 5, and reported ACC on the Scene15 and CUB datasets, as shown in Figs. 3 and 4.

For the Scene15 dataset (Fig. 3(a)), the model achieves the highest ACC when  $k = 25$ , suggesting that more neighborhood helps to better exploit sample structure. However, the smallest interquartile range

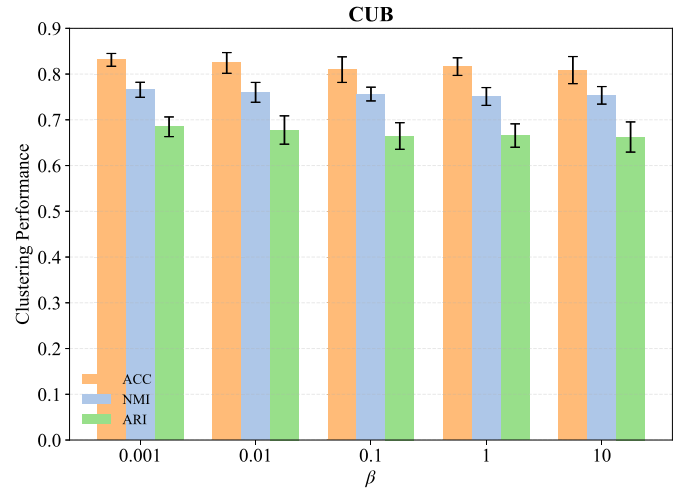
(IQR) is observed at  $k = 10$ , indicating a more stable performance. In contrast, the results at  $k = 15$  and  $k = 20$  exhibit higher variability and lower median, implying potential instability and performance degradation at intermediate values. For the CUB dataset (Fig. 3(b)), the highest ACC is obtained at  $k = 10$ , with the smallest variance, indicating that a moderate neighborhood size provides both robust and effective clustering. Larger values of  $k$ , such as 20 and 25, do not consistently improve performance and introduce higher fluctuations. This suggests that including too many neighbors may introduce noisy, especially in datasets like CUB.

Building on the sensitivity trends observed above, Fig. 4 further illustrates the performance gap between the SRLS model and the isolated complementarity-aware module ( $\beta = 0$ ). It is observed that the performance of the complementarity-aware module (blue bars) remains remarkably stable as  $k$  varies from 5 to 25. Specifically, on the Scene15 dataset, the two trend lines remain nearly parallel, indicating that the feature reconstruction module provides a stable and reliable performance boost regardless of the neighborhood scale. On the CUB dataset, while the ablated module is robust to neighbor selection, the synergistic effect is most pronounced at  $k = 10$ , where the full model maximizes the benefit of an optimal neighborhood. This robustness demonstrates that our heterogeneous graph propagation can effectively capture complementary signals across a wide range of neighborhood configurations.

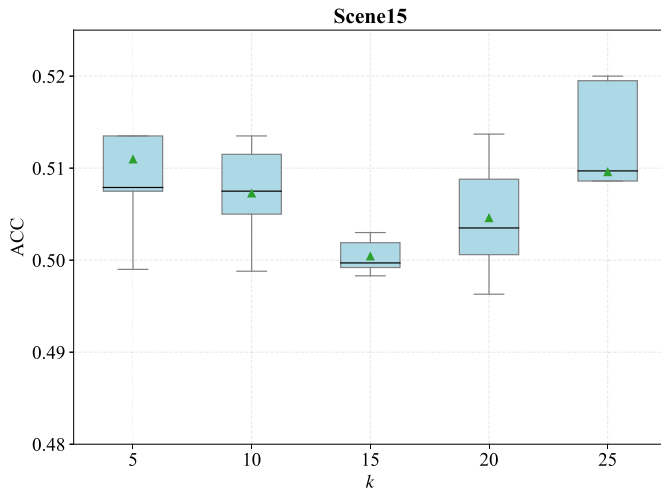
In summary,  $k = 10$  is empirically chosen as it balances precision and robustness while ensuring stable performance across most datasets.



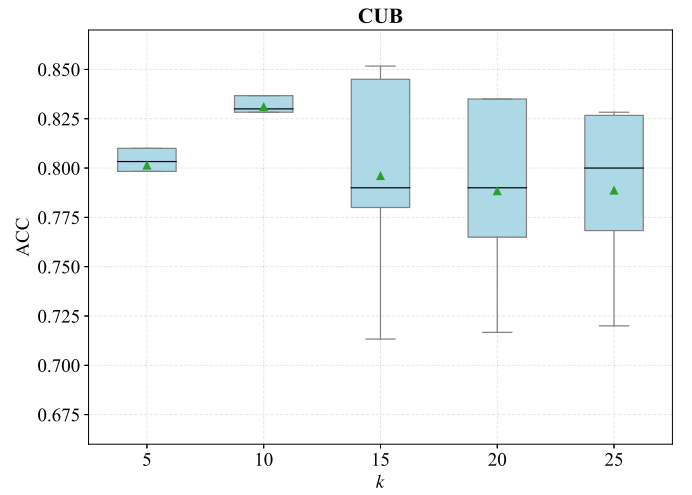
(a) Scene15



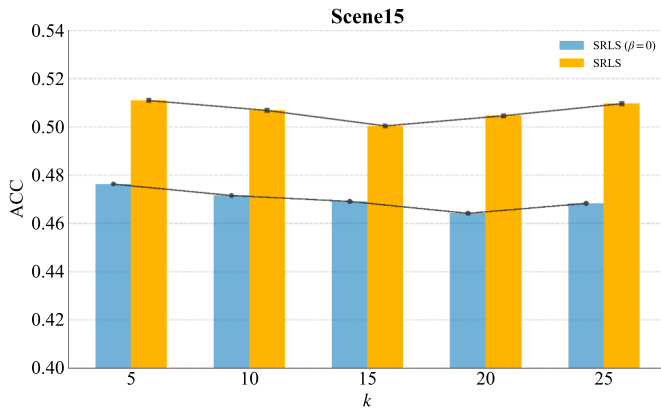
(b) CUB

Fig. 2. The influence of  $\beta$  on Scene15 and CUB datasets.

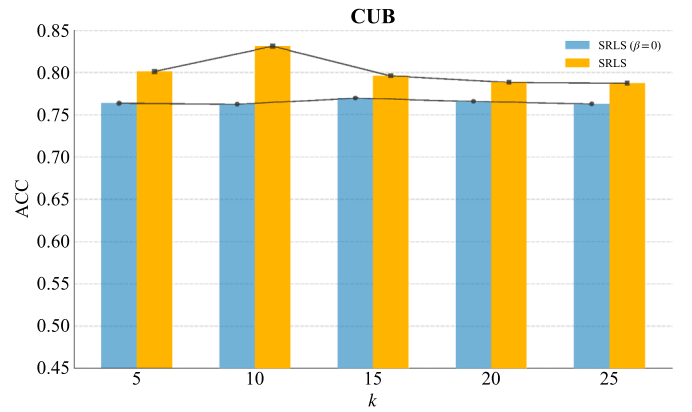
(a) Scene15



(b) CUB

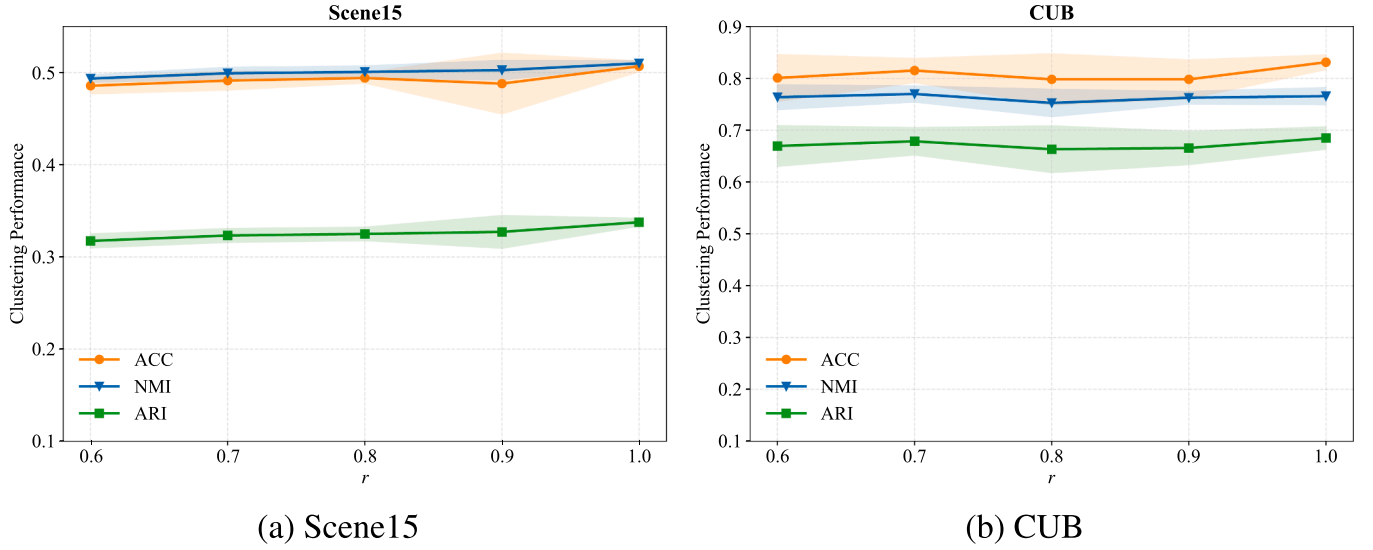
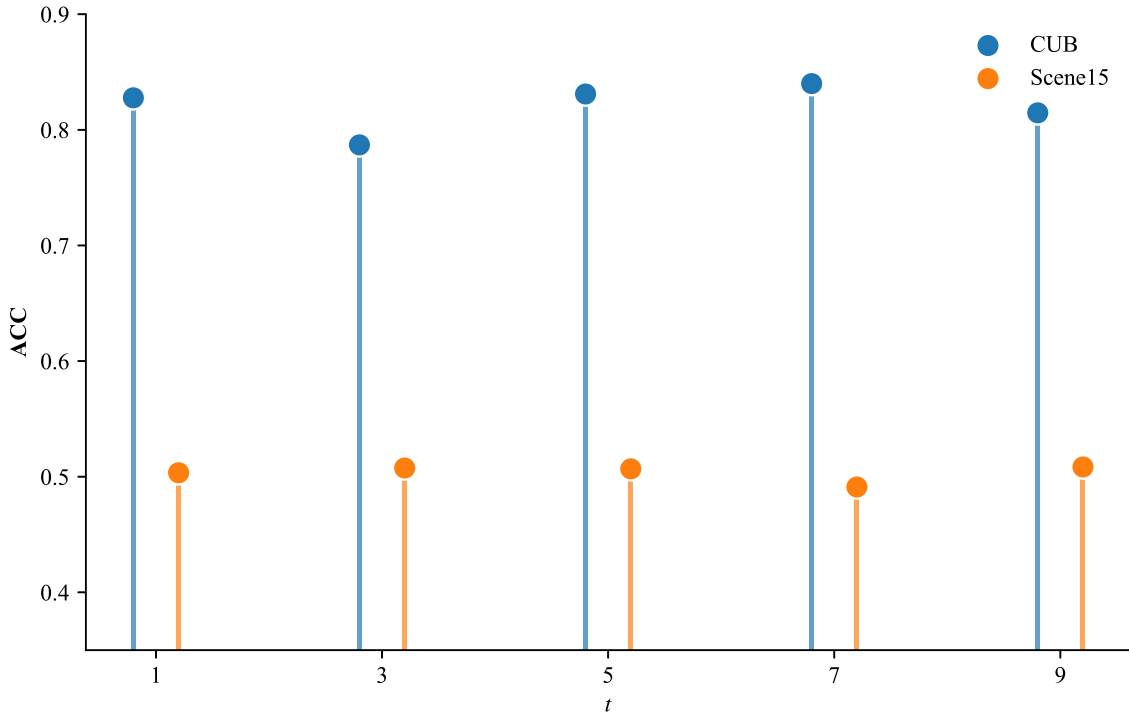
Fig. 3. Overall performance and robustness of SRLS to  $k$  on Scene15 and CUB.

(a) Scene15



(b) CUB

Fig. 4. The influence of  $k$  on Scene15 and CUB datasets.

Fig. 5. The influence of  $r$  on Scene15 and CUB datasets.Fig. 6. The influence of  $t$  on Scene15 and CUB datasets.

**Influence of feature reconstruction ratio  $r$ .** In the consistency-guided feature reconstruction module, the hyperparameter  $r$  controls the ratio of samples within each batch that are used to optimize the loss of feature reconstruction. As shown in Fig. 5, we evaluate the clustering performance under varying values of  $r \in \{0.6, 0.7, 0.8, 0.9, 1.0\}$ .

On the Scene15 dataset (Fig. 5(a)), the clustering performance remains stable across different values of  $r$ , with an improvement observed at  $r = 1.0$ . Similarly, on the CUB dataset (Fig. 5(b)), ACC and NMI show a slight upward trend as  $r$  increases, while ARI remains nearly constant. These observations suggest that increasing the value of  $r$  can contribute to gains in clustering quality without causing instability. Overall, the results indicate that the model is relatively insensitive to the choice of  $r$ , and using a larger selection ratio ensures full utilization of training information. Therefore, we set  $r = 1.0$  in all experiments to achieve optimal and stable performance.

**Influence of  $t$ .** Fig. 6 evaluates the influence of the  $t$ -order matrix by varying  $t \in \{1, 3, 5, 7, 9\}$ . On the CUB dataset, the performance peaks at  $t = 7$ , demonstrating the clear benefit of aggregating higher-order information for complex data. Scene15 exhibits high robustness across all tested values. We empirically select  $t = 5$  as the default setting for all experiments. This choice strikes an optimal balance between complementarity aggregation and computational efficiency, ensuring stable and competitive performance across diverse benchmarks without over-fitting to a specific dataset.

**Influence of the number of propagation layers  $L$ .** We investigate the sensitivity of SRLS to the number of propagation layers  $L \in \{1, 2, 3, 4\}$ . Fig. 7 illustrates the relative gain in ACC across five benchmark datasets, where the performance of  $L = 1$  serves as the zero-gain baseline.



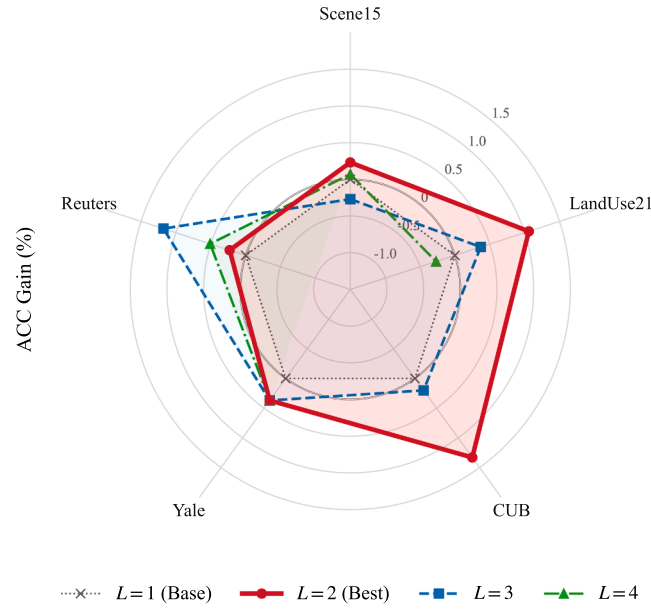


Fig. 7. Relative performance gain comparison across different propagation layers  $L$ .

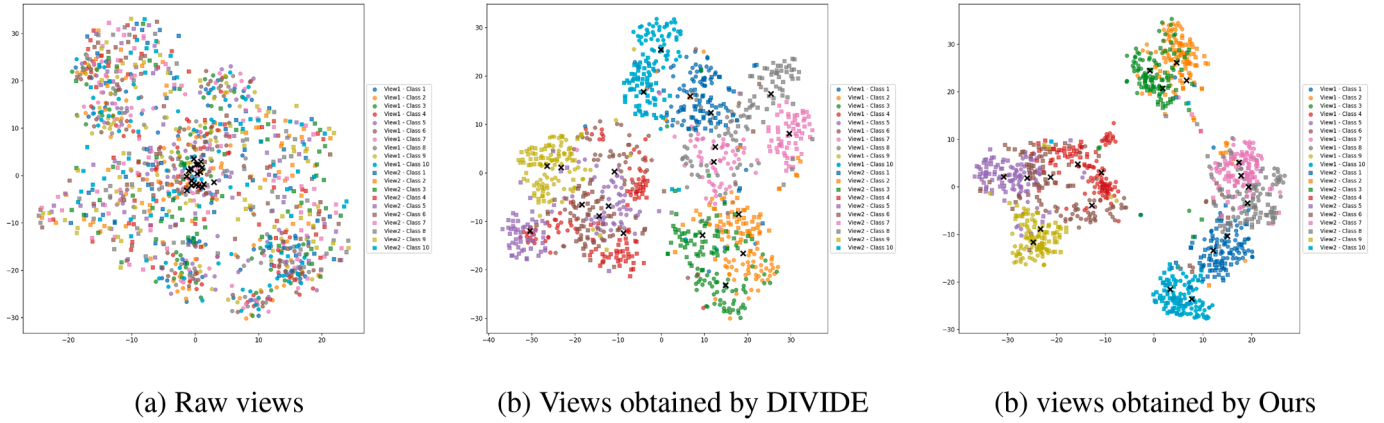


Fig. 8. Comparison of different views on CUB dataset.

It is observed that  $L = 2$  consistently achieves the highest relative gains across all datasets, particularly on *CUB* and *LandUse21*. This emphasizes the importance of aggregating second-order neighborhood information to capture intrinsic structures and refine cluster boundaries. Conversely, performance declines when  $L \geq 3$  due to the over-smoothing effect inherent in graph neural networks, which makes node representations less distinguishable and obscures discriminative features. Consequently,  $L = 2$  is selected as the optimal configuration to balance structural richness and representation stability.

#### 4.5. Visualizations (Q4)

Fig. 8 consists of three scatter-plot subfigures, each depicting two views (View 1: circles; View 2: squares) of the same set of instance belonging to ten distinct classes (different colors). In each subfigure, “x” markers indicate the centroid (mean) of each class in the embedding space of each view.

In the raw view representations (Fig. 8(a)), clusters are heavily intermingled. In both View 1 and View 2, samples from different clusters occupy overlapping regions, which implies that partitions based solely on Euclidean distance would be highly unreliable. Furthermore, for any given clusters, the centroids computed in View 1 and View 2 are far

apart, indicating that each view distribution has its own mean and does not align in the shared 2D projection.

After applying DIVIDE (Fig. 8(b)), clusters become more compact and their centroids shift closer across views, showing that some alignment has occurred. However, centroids remain misaligned and clusters still overlap in places, and intra-cluster spreads are uneven—indicating that DIVIDE does not fully reconcile distribution differences between views or achieve uniform cluster structure.

Through proposed model (Fig. 8(c)), clusters form tight, well-separated clusters in both views with minimal overlap. The centroids of each class coincide on View 1 and View 2, confirming the exact cross-view alignment. Clusters are uniform in shape and scale, reflecting regularized intra-class variance. The distances between classes are balanced, maximizing the margins for downstream clustering. Very few outliers remain, showing that our model produces robust, discriminative representations.

Similarly at the sample level, in the raw samples (Fig. 9(a)) clusters are diffuse and classes overlap heavily, centroids are scattered which indicates poor separability. After applying DIVIDE (Fig. 9 (b),  $NMI = 0.7225$ ), each class becomes more compact and centroids shift closer across views, but overlap and uneven intraclass spreads remain, which show only partial alignment. In contrast, our model (Fig. 9(c),  $NMI = 0.7745$ ) produces tight well-separated clusters whose centroids coincide

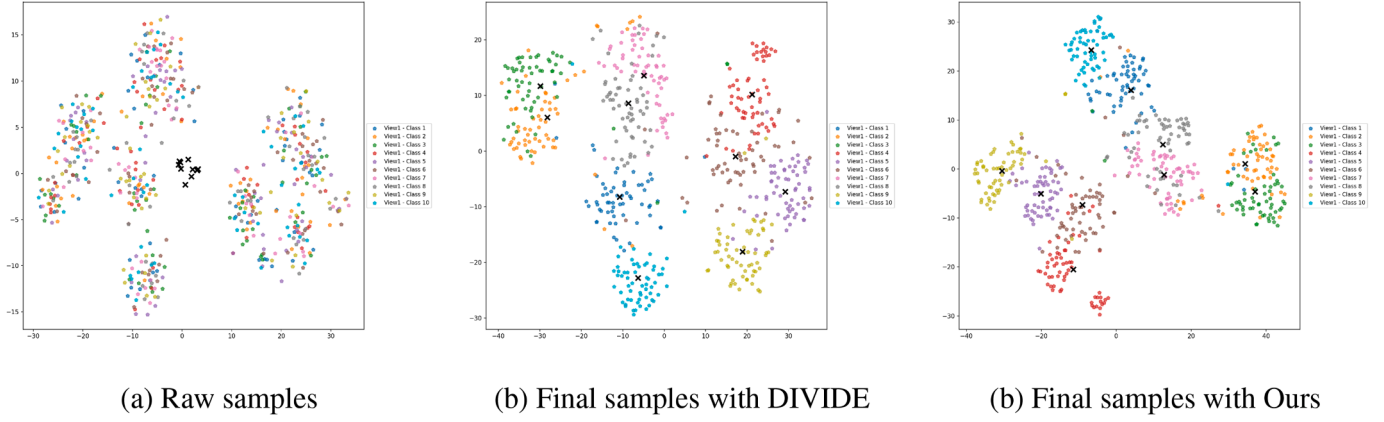


Fig. 9. Comparison of sample structure.

Table 5

Comparison of performance across various datasets under different missing rates.

| Missing Rate | Method | Scene15 |        |        | LandUse21 |        |        | CUB    |        |        |
|--------------|--------|---------|--------|--------|-----------|--------|--------|--------|--------|--------|
|              |        | ACC     | NMI    | ARI    | ACC       | NMI    | ARI    | ACC    | NMI    | ARI    |
| 0.1          | DIVIDE | 0.4605  | 0.4486 | 0.2830 | 0.3081    | 0.3573 | 0.1609 | 0.7727 | 0.7203 | 0.6166 |
|              | Ours   | 0.4788  | 0.4823 | 0.3082 | 0.3105    | 0.3623 | 0.1673 | 0.7973 | 0.7426 | 0.6503 |
| 0.3          | DIVIDE | 0.4598  | 0.4487 | 0.2834 | 0.3030    | 0.3566 | 0.1582 | 0.7570 | 0.7050 | 0.5897 |
|              | Ours   | 0.4891  | 0.4778 | 0.3127 | 0.3015    | 0.3471 | 0.1573 | 0.7757 | 0.7316 | 0.6287 |
| 0.5          | DIVIDE | 0.4430  | 0.4253 | 0.2620 | 0.2999    | 0.3550 | 0.1584 | 0.6983 | 0.6527 | 0.5300 |
|              | Ours   | 0.4503  | 0.4525 | 0.2827 | 0.3012    | 0.3498 | 0.1585 | 0.7207 | 0.6792 | 0.5643 |
| 0.7          | DIVIDE | 0.4246  | 0.4097 | 0.2439 | 0.2924    | 0.3411 | 0.1531 | 0.6797 | 0.6251 | 0.5011 |
|              | Ours   | 0.4104  | 0.4100 | 0.2430 | 0.2905    | 0.3386 | 0.1553 | 0.6950 | 0.6452 | 0.5198 |

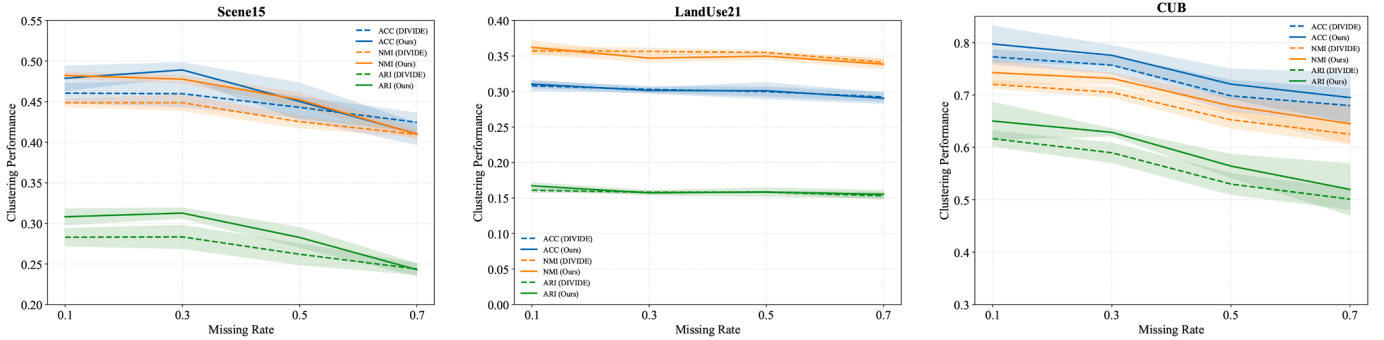


Fig. 10. Performance comparison under different missing rates on different datasets.

across views, the clusters have uniform shape and scale and the outliers are minimal, demonstrating superior alignment of the cross-view and discriminability at the sample level.

#### 4.6. Performance under different missing rates (Q5)

To investigate the robustness of our proposed model, we conduct experiments by changing the missing rates. To be specific, we vary the missing rate  $\eta$  from 10% to 70% with a gap of 20% on different datasets. As shown in Table 5, performance for both methods declines as the missing rate increases. However, SRLS consistently outperforms DIVIDE, particularly on the CUB dataset, demonstrating superior robustness. The lead in NMI and ARI highlights our model's ability to preserve cluster structures despite missing views. This advantage stems from the dual interaction mechanism: complementarity-aware propagation compensates for missing information, while consistency-guided reconstruction maintains feature discriminability, making SRLS more effective in recovering semantics from sparse data.

In addition, we supplement the corresponding trend and standard deviations in Fig. 10. The shaded areas around the curves represent the standard deviation of multiple runs, reflecting the stability of model performance. It can be observed that while both models exhibit fluctuations as the missing rate increases, our method consistently outperforms DIVIDE across all metrics (ACC, NMI, and ARI) in nearly all scenarios. Furthermore, our model shows better stability, as indicated by generally smaller standard deviations. Among the metrics, ACC and ARI exhibit more significant performance degradation as the missing rate increases, suggesting that they are more sensitive to information loss. In contrast, NMI remains relatively stable, possibly due to its robustness in capturing the overall structure of the cluster distribution. The most notable differences occur at low missing rates (e.g., 0.1 and 0.3), where our proposed model shows the most significant advantage. This implies that when sufficient information is available, the proposed model can better capture complementary information and perform more accurate feature reconstruction, thereby benefiting downstream clustering tasks. It is noteworthy that as the missing rate increases to extreme levels (e.g.,

**Table 6**

Comparison of total runtime (s) and peak GPU memory usage (MB).

| Method | Metric   | Scene15 | LU21   | CUB    | Yale   | Reuters |
|--------|----------|---------|--------|--------|--------|---------|
| CVCL   | Time (s) | 346.33  | 98.01  | 42.60  | 41.31  | 879.90  |
|        | Mem (MB) | 392.80  | 202.90 | 306.90 | 192.90 | 202.60  |
| MFLVC  | Time (s) | 558.14  | 470.70 | 329.70 | 48.47  | 3940.14 |
|        | Mem (MB) | 397.00  | 267.90 | 201.20 | 294.90 | 1192.60 |
| GCFAGG | Time (s) | 729.96  | 397.11 | 68.51  | 5.87   | 2751.06 |
|        | Mem (MB) | 290.40  | 290.70 | 313.40 | 329.30 | 289.30  |
| DIVIDE | Time (s) | 295.00  | 367.70 | 203.66 | 85.96  | 728.33  |
|        | Mem (MB) | 215.70  | 206.70 | 183.30 | 363.00 | 193.70  |
| Ours   | Time (s) | 350.04  | 322.98 | 211.80 | 98.96  | 460.29  |
|        | Mem (MB) | 222.20  | 221.10 | 218.90 | 369.80 | 399.70  |

0.7), the performance gap between SRLS and DIVIDE tends to narrow. This phenomenon of performance convergence aligns with expectations, as extreme information scarcity constrains the capacity of any model to extract discriminative cluster structures. Nevertheless, SRLS maintains a consistent lead even under such adverse conditions, further validating the resilience of our sample interaction mechanism.

#### 4.7. Efficiency analysis (Q6)

Table 6 compares the efficiency of SRLS against representative consistency-based (CVCL, MFLVC) and complementarity-based (GCFAGG, DIVIDE) methods. We emphasize peak GPU memory as the primary hardware bottleneck. As illustrated, SRLS demonstrates superior memory stability and scalability. On the large-scale Reuters dataset, while MFLVC's memory usage escalates to 1192.60 MB, SRLS maintains a significantly lower footprint of 399.70 MB. This indicates that our synergistic mechanism is more resource-efficient than complex multi-level contrastive or global graph-based methods. Regarding training speed, SRLS achieves a runtime comparable to DIVIDE across most datasets, ensuring high efficiency without significant temporal overhead. In summary, SRLS provides a balance between clustering performance and hardware resource consumption, making it highly suitable for memory-constrained scenarios.

## 5. Conclusion

In this paper, we proposed SRLS, which leverages dual sample interactions for multi-view clustering. The core strength of our model is the systematic integration of inter-sample complementarity and intra-sample consistency, resulting in more discriminative representations. We believe this dual-modeling paradigm provides a valuable template to balance different multi-view characteristics in self-supervised learning. While SRLS achieves superior clustering performance and robustness across various benchmarks, its reliance on empirical hyperparameter tuning remains a limitation.

Future work will focus on two parts. First, we will explore parameter-free mechanisms and joint end-to-end optimization for clustering to reduce manual tuning. Second, we plan to enhance the model's scalability for massive-scale datasets to broaden its practical applicability in real-world scenarios.

#### CRedit authorship contribution statement

**Ruina Bai:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Funding acquisition, Formal analysis; **Ruizhang Huang:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization; **Yongbin Qin:** Supervision, Software, Resources, Funding acquisition; **Jingjing Xue:** Visualization, Data curation; **Rujun Tian:** Visualization, Data curation.

#### Data availability

Data will be made available on request.

#### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Acknowledgment

This work was supported by the [National Key R&D Program of China](#), No. 2023YFC3304500, Guizhou Provincial Science and Technology Department Project, No. Qiankehe MS[2026] 077 and Natural Science Fund of GZU, No. (2024)31.

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